

Where do the vectors live?

pgvector — inside the Postgres we already run — versus Chroma DB as a new, dedicated vector store. One decision, and it turns on a single fact: we handle **PHI**.

Recommendation → pgvector

Keep PHI in the datastore already inside our compliance boundary.

We are a HIPAA covered entity

ALREADY TRUE

Postgres 15 holds PHI

- ✓ System of record for patient data
- ✓ Already under our encryption, access & audit posture
- ✓ On managed RDS / Aurora → HIPAA-eligible under the AWS BAA

THE BAR ANY OPTION MUST CLEAR

No new PHI path without a BAA

- ✗ Don't ship PHI to a vendor with no BAA
- ✗ No new multi-tenant residency surprise
- ▲ A new system means new ops, review & cost

Verdict matrix

CRITERION	PGVECTOR	CHROMA DB
HIPAA / BAA	Inherits Postgres AWS BAA on RDS / Aurora	None advertised SOC 2 Type II only
Data residency	Existing account No egress	Multi-tenant cloud us-east-1 / eu-west1
New vendor?	No	Yes
Integration cost	CREATE EXTENSION One column, SQL joins	New client + sync Embeddings leave Postgres
Ops burden	Reuse Postgres ops	New system / vendor
Scale ceiling	Single-node HNSW Fine to millions	Distributed / serverless Wins at billions

Chroma leads on raw scale only – which we don't need.

Compliance posture

AWS BAA

pgvector inherits Postgres's HIPAA eligibility. PHI never leaves our boundary.

✓ BAA available

SOC 2 only

No HIPAA claim on Chroma's pricing, security, or docs pages. No BAA offered.

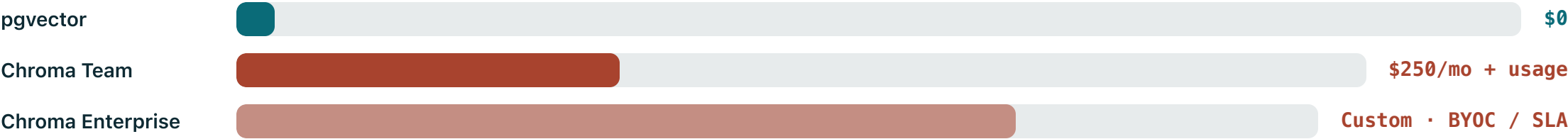
× No BAA

Routing patient records through managed Chroma sends PHI to a third-party multi-tenant service with no business-associate contract — a HIPAA violation. BYOC (Enterprise, custom) could keep data in-VPC, but still carries no advertised BAA.

pgvector, in our existing Postgres

✓ pgvector

No new vendor. No new data path. No new BAA. One extension.



The extension is free — the cost is the Postgres we already pay for. Chroma usage adds write \$2.50/GiB, storage \$0.33/GiB·mo, egress \$0.09/GiB on top of the plan. The cheapest option is also the compliant one.

When Chroma wins

CHROMA'S REAL STRENGTHS

- ✓ Corpus at 100M–billions of vectors
- ✓ High write / query throughput
- ✓ Purpose-built metadata filtering & tooling
- ✓ Serverless object-storage scaling

...BUT ONLY WHEN

- ▲ The corpus is **non-PHI** (SOC 2 suffices), or
- ▲ You run **BYOC** in your own VPC under Enterprise
- ✗ Neither applies to Riverbend today